# FINITE LIMITS FROM FINITE PRODUCTS, EQUALIZERS, AND A FINAL OBJECT

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*Editorial note.* Counted span is the author's own proof environment, source lines 2275-2288, following the statement at 2264-2273.

*Editorial note.* The proof is complete; the displayed equalizer identification carries a routine universal-property check recorded as an obligation.

*Allowed standard prerequisites (editorial).*

- Limits from products and equalizers. lemma-limits-products-equalizers, cited at proof lines 2278-2280. Supplies that existence of finite products and equalizers implies existence of finite limits.
- Finite limits include finite products, fibre products, equalizers, final objects. the limit definitions recalled at proof lines 2276-2278. Supplies the instantiation direction (1) implies (2) and (3).

*Source reference.* Original lemma 002N: sources/stacks-categories/pinned-original-source.tex, source lines 1745–1745. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $\mathcal{C}$ be a category. The following are equivalent:*

(1) *Finite limits exist in $\mathcal{C}$.*
(2) *Finite products and equalizers exist.*
(3) *The category has a final object and fibre products exist.*

*Proof.* Since finite products, fibre products, equalizers, and final objects are limits over finite index categories we see that (1) implies both (2) and (3). By Lemma 002N above we see that (2) implies (1). Assume (3). Note that the product $A \times B$ is the fibre product over the final object. If $a, b : A \to B$ are morphisms of $\mathcal{C}$, then the equalizer of $a, b$ is

$$(A \times_{a,B,b} A) \times_{(\mathrm{pr}_1,\mathrm{pr}_2),A\times A,\Delta} A.$$

Thus (3) implies (2) and the lemma is proved. □

*Date*: stacks-project:categories.tex:local-snapshot-20261012; GNU FDL 1.2 or later.